Content

Learning objectives and content

Objectives
After completing the course, the student will be able to

- Explain how open-loop transfer function properties can be modified using feedback
- Understand the relevance of frequency-domain control-design methods and represent a frequency-response function using a Bode plot
- Assess closed-loop stability and its robustness using the Nyquist plot of the open-loop frequency-response function
- Design single-input-single-output feedback controllers by shaping the open-loop frequency-response function
- Identify the limitations of linear control design, as dictated by the Bode Sensitivity integral and the inherent trade-offs between tracking performance and disturbance rejection, and sensitivity to sensor noise
- Understand the relevance of state-space control-design methods and analyze the state-space equations
- Design feedback controllers using state-space design techniques by designing state-feedback controllers and state estimators using pole-placement technique
- Understand the limitations of controller design due to sampling and the limitation of the experimental platform.

Content
Control systems are omnipresent in everyday use of technology. The basic principles of control of linear dynamic systems are explored, and design methods are treated for designing control systems to handle design specifications in terms of speed (bandwidth), stability and robustness under presence of disturbances and sensor noises. Achievements and limitations are being addressed and the theory is complemented with design assignments on laboratory setups.

2026

Control systems (5ESH0)

Practical information

Course module
5ESH0

Course type
Bachelor College

Credits (ECTS)
5 EC

Category
Deepening

Language of instruction
English

Offered by
Electrical Engineering - Control Systems

Course enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 4
  Timeslot(s)
  E - Mo 9-10, Tu 5-8, Th 1-4

  Enrolment period
  15 November 2026 until 21 March 2027 23:59

  Unenrolment
  until 19 April 2027 23:59

Entrance requirements

Assumed previous knowledge
5ESF0, Signals and systems
2WBB0, Calculus
5EZA0, Math 1

Entrance requirements course enrolment
You must meet the following requirements

- In Block 4 you may not be part of one of the following target groups
  - Bachelor Automotive Technology before revision
  - Bachelor Electrical Engineering before revision

Instructional modes

Instructional modes

- Guided selfstudy
  Remark
- Lecture
  Remark

Tests

Tests

- Assignment
  Interim examination

  Test weight
  20

  Minimum grade
- Written examination
  Final examination

  Test weight
  80

  Minimum grade
  5.0

Test enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Assignment

- Block 4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

Written examination

- Block 4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

  Unenrolment
  until 6 June 2027

  Test date
  29 June 2027

- Block I
  Enrolment period
  15 November 2026 until 1 August 2027 23:59

  Unenrolment
  until 1 August 2027

  Test date
  17 August 2027

Lecturer(s)

Co-lecturer

- B. Terzin
- G. Mitsos
- J.S. Petrulis
- K. Wang
- T. Bouzid

Contactperson for the course

- dr.ir. M.C.F. Donkers

Responsible lecturer

- dr.ir. M.C.F. Donkers

Subject matter expert

- dr.ir. M.C.F. Donkers

Materials

Recommended material

- Feedback Control of Dynamic Systems, by Franklin, Powell and Emami-Naein.
- Other online material available via CANVAS

Do you want to know more?

dr.ir. M.C.F. Donkers can tell you more about this course.
Send an e-mail to M.C.F.Donkers@tue.nl